



Reducing Antisolvent Use in the STRAP Process by Enabling a Temperature-Controlled Polymer Dissolution and Precipitation for the Recycling of Multilayer Plastic Films

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The recently reported processing strategy called solvent-targeted recovery and precipitation (STRAP) enables deconstruction of multilayer plastic packaging films into their constituent resins by selective dissolution. It uses a series of solvent washes that are guided by thermodynamic calculations of polymer solubility. In this work, the use of antisolvents in the STRAP process was reduced and solvent mixtures were considered to enable the temperature-controlled dissolution and precipitation of the target polymers in multilayer films. This was considered as a means to further improve the STRAP process and its estimated costs. Two STRAP approaches were compared based on different polymer precipitation techniques: precipitation by the addition of an antisolvent (STRAP-A) and precipitation by decreasing the solvent temperature (STRAP-B). Both approaches were able to separate the constituent

polymers in a post-industrial film composed primarily of polyethylene (PE), ethylene vinyl alcohol (EVOH), and polyethylene terephthalate (PET) with near 100% material efficiency. Technoeconomic analysis indicates that the minimum selling price (MSP) of the recycled resins with STRAP-B is 21.0% lower than that achieved with STRAP-A. This provides evidence that thermally driven polymer precipitation is an option to reduce the use of antisolvents, making the STRAP process more economically and environmentally attractive. A third process, STRAP-C, was demonstrated with another post-industrial multilayer film of a different composition. The results demonstrate that this process can also recover polymers at similar costs to those of virgin resins, indicating that the STRAP technology is flexible and can remain economically competitive as the plastic feed complexity is increased.

Introduction

One of the largest sectors for single-use plastics is the packaging industry, which accounted for over 35% of the plastics produced in 2015.^[1] Flexible plastic packaging has the advantages of protecting enclosed products from breakages or contamination, extending shelf life, providing an attractive appearance, and reducing packaging size and weight.^[2] These plastic packaging materials can come in the form of multilayer films, which are composites of distinct polymers that are combined to achieve specific properties that cannot be

provided by single plastic layers.^[3] Each layer gives the final bulk plastic a property advantage that is desirable for food packaging, medical packaging, and other applications. It is the different layers that provide sealing properties, oxygen and water barriers, and structural rigidity.^[4] In most cases, these multilayer materials are intended for single use, as there is currently no technology available for their widespread economical recycling. Specifically, traditional recycling methods such as mechanical recycling cannot process these materials;^[2,5] this is because the different polymers are not chemically compatible and will phase separate during mechanical recycling.^[6,7] Multilayer plastic materials can consist of more than 17 layers of different polymers, which makes it challenging to recover all of the components.^[8–10] Further technology development is needed for the more efficient collection, sorting, separation, and recycling of these multilayer packaging materials, as they represent an increasing amount of post-consumer (PCW) and post-industrial plastic waste (PIW).

Our team has recently reported a new processing method for the recycling of multilayer plastic films into pure resins by selectively dissolving each polymer into a different solvent in an approach we call solvent-targeted recovery and precipitation (STRAP).^[11] Solvent-based plastics recycling approaches have been studied by several researchers and are currently being implemented by a number of companies.^[2,12–17] For example, APK AG's Newcycling process is being carried out at a capacity of 8000 metric tons per year in Germany for the recycling of

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polyethylene (PE) and polyamides (PA) from multilayer films.^[18,19] Another process is the CreaSolv process developed by CreaCycle GmbH and the Fraunhofer Institute IVV, which targets thermoplastics from plastic packaging waste, electronic waste, and construction insulation foam.^[19,20] One of their plants has a capacity of 2000–4000 tons per year. Most of the solvent-based processes in the industry are in the early commercial or pilot stage of maturity, demonstrating that there is interest in implementing these technologies. In our case we used the STRAP process to convert a commercially available post-industrial rigid multilayer film (A1) manufactured by Amcor into its three main polymer components with 100% recovery. This multilayer film consisted primarily of PE, ethylene vinyl alcohol (EVOH), and polyethylene terephthalate (PET), with various tie layers that include ethylene vinyl acetate (EVA). In this process, the film is dissolved in two different solvents at a high temperature, each of which solubilizes an individual polymer. Antisolvents are then added to precipitate the dissolved polymer from the solvent. The solvent selection for STRAP is based on quantifying polymer and solvent thermodynamic properties using three computational methods of increasing accuracy and complexity: the calculation of Hansen solubility parameters (HSPs), molecular dynamics (MD) simulations, and a combined quantum chemical and statistical mechanical approach called the conductor-like screening model for realistic solvents (COSMO-RS). In general terms, HSPs are used for the rapid screening of solvents suitable for the selective dissolution of specific polymers, MD simulations provide detailed calculations of polymer structures and conformations, and COSMO-RS enables predictions of polymer solubility as a function of both the temperature and the liquid phase composition using the conformations from MD. A rigorous technoeconomic analysis (TEA) indicated that STRAP can produce the recycled resins at lower costs than the virgin resins when the process operates at a size greater than 3000 tons per year.^[11] However, the TEA also showed that the largest costs were associated with the thermal separation (distillation) of the solvents and antisolvents, accounting for 33.6 and 79.3% of the capital and operating costs, respectively. It would thus be highly desirable to reduce the costs of the STRAP technology and it would be helpful to see if STRAP can be applied to other types of multilayer packaging materials.

The objective of this paper is to identify operating conditions that will reduce the amount of antisolvents used in the STRAP process and to demonstrate how STRAP can be applied to a more complex, industrial multilayer film (provided by Amcor). We combine thermodynamic computational tools, polymer characterization methods, and modern process design tools to develop a realistic near-term approach to design solvent systems to recycle multilayer plastics. The efficient utilization of these solvents requires a process that can recover and reuse the solvents, while minimizing the amount of solvents left as waste material. An approach to reduce the use of antisolvents is to use solvents or solvent mixtures that can enable the temperature-controlled dissolution and precipitation of the target polymers in a multilayer film. The use of solvent combinations has been considered as an option in dissolution/

precipitation methods for certain polymers. Some studies have used solvent mixtures only for the dissolution of the target plastic to later add an antisolvent for precipitation. Hadi et al. studied mixtures of turpentine/petroleum ether (PetE) to dissolve polyolefins and later precipitate them using *n*-hexane and PetE.^[21] Weeden et al. used mixtures of heptane/dichloromethane to dissolve electronic waste components.^[22] However, they recovered the polymers by evaporating the solvents. Pure solvents and solvent combinations have also been used to dissolve various polymers at different temperatures, reducing the number of solvents needed to recover each polymer. Some patents disclosing this technique have also relied on evaporating the solvent to obtain the final solid polymers.^[23,24] Precipitating the polymers through a solvent temperature change instead of evaporating the solvent or adding an antisolvent could be a way to make dissolution/precipitation methods more feasible.^[15]

In this paper, we evaluate and compare the STRAP technology using two different polymer precipitation techniques: precipitation by the addition of an antisolvent (STRAP-A) and precipitation by decreasing the solvent temperature (STRAP-B). Two different post-industrial rigid multilayer films manufactured by Amcor (A1 and A2) were used to demonstrate the STRAP process. The multilayer film A1 was composed of PE, EVOH, and PET with minor component EVA. Film A2 was composed of the same polymers with the addition of glycol-modified polyethylene terephthalate (PETG). Thermodynamic tools were used for solvent selection and the experimental results were used in a process model that allowed an estimation of the process costs.

Computational modeling for solvent screening

We assessed polymer solubility computationally to guide solvent selection for the STRAP process; our approach uses a combination of HSPs, MD simulations, and COSMO-RS, following our previous work.^[11] HSPs are empirical solvent parameters that are widely used to qualitatively identify solvents that are capable of dissolving target polymers.^[25] Each compound (polymer or solvent) is assigned three parameters that account for dispersion, polar, and hydrogen-bonding forces.^[26] These three parameters are used as coordinates that locate the compounds in HSP space. Each polymer has an additional radius parameter, R_0 , that defines a sphere in HSP space. Solvents with HSPs within this sphere are expected to dissolve the polymer, whereas solvents with HSPs outside of this sphere are not expected to dissolve the polymer. HSPs (and values of R_0) have been tabulated for a wide range of polymers and solvents based on solubility experiments in reference solvent systems that span the HSP space.^[25] In this work, HSPs were obtained from the HSP handbook^[26] and used for preliminary solvent selection, although these values do not provide quantitative predictions of solubility.

MD simulations and COSMO-RS are used for further quantitative solubility predictions utilizing an updated protocol compared to our previous work. Atomistic MD simulations for

oligomers were performed to obtain input structures for COSMO-RS. MD simulations were first performed of a single oligomer in dilute solution in the isothermal-isobaric ensemble. Each MD simulation contained one oligomer molecule (EVOH or PETG) and 216 solvent molecules. EVOH was simulated in water and PETG in toluene. The EVOH oligomer molecule contained four vinyl alcohol repeat units and two ethylene repeat units. The PETG oligomer contained four ethylene glycol repeat units and two 1,4-cyclohexanedimethanol repeat units. The radius of gyration (R_g) and solvent accessible surface area (SASA) of the oligomer were then calculated as a function of simulation time from the MD trajectories to characterize the distribution of oligomer conformations. Representative oligomer conformations that span the R_g -SASA space were extracted from the MD trajectories and used as input to the COSMO-RS workflow detailed below. A total of 22 conformers were taken from the EVOH trajectory and 18 conformers were taken from the PETG trajectory. MD simulations were performed using Gromacs 2016.^[27] All compounds were parameterized by using Antechamber and the Generalized AMBER force fields.^[28,29]

COSMO-RS predicts the equilibrium properties of multi-component systems based on quantum mechanical calculations and statistical thermodynamics methods.^[30] It represents each molecule based on the screening charge density that arises at its molecular surface due to the polarization of the medium. The screening charge density is obtained from density functional theory (DFT) calculations and histogrammed to generate

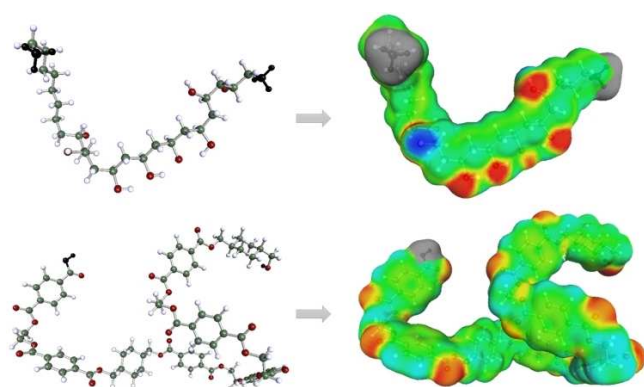


Figure 1. Molecular structures and corresponding COSMO-RS screening charge densities of an example EVOH oligomer (top) and PETG oligomer (bottom). Regions with negative screening charge densities are in blue, positive screening charge densities are in red, intermediate values are in green and teal, and deactivated groups (which do not contribute to the σ -profile) are in grey.

a σ -profile. We approximated the σ -profile of a polymer by generating σ -profiles for oligomer structures with deactivated terminal groups.^[31] The screening charge density and resulting σ -profile depends on the specific molecular conformation; consequently, we generated a set of oligomer conformations using MD (as described above) to span a range of possible σ -profiles. Figure 1 provides an example of two conformers for different oligomers and their DFT calculation results of screening charge distributions. The chemical potential of a polymer is calculated based on the σ -profiles to enable predictions of polymer solubility via a solid-liquid equilibrium calculation. DFT calculations of screening charge distributions were performed by Gaussian 16 at the BVP86/TZVP/DGA1 level of theory.^[32] COSMO-RS solubility calculations were performed by COSMOtherm 19 with the parameterization BP_TZVP_19.^[30,33]

Results and Discussion

STRAP-A with multilayer film A1

The STRAP-A process followed the same procedure as in our previously published work and it was used as a benchmark to compare with the alternative processes in this study. Table 1 shows that an overall average mass balance of 102.58 wt% was obtained with a standard deviation of ± 0.96 wt% for the recovery of PE, PET, and EVOH from multilayer film A1. The overall mass balances exceeded 100 wt% possibly due to solvent uptake. Drying conditions will need to be adjusted to minimize entrained solvents. The values reported in Table 1 are consistent with our previously published results.^[11] Figure 2 shows solvents, antisolvents, temperatures and dissolution times that were used in each process step for STRAP-A. Acetone and water were used as antisolvents for the precipitation of PE and EVOH, respectively.

The extensive use of antisolvents can be a concern at larger scales due to potential high costs and environmental and safety risks. It was determined that the energy required to separate the respective solvents and antisolvents used in STRAP-A accounted for 33.6% of the total capital costs and 79.3% of the total operating costs.^[11] The distillation columns in STRAP-A were the main cost drivers in the process. An alternative to using antisolvents for polymer precipitation is to cool the solvent with the dissolved polymer to induce the precipitation. For this purpose, solvent mixtures were considered to both dissolve the target polymer at a high temperature and to easily

Table 1. Yield of polymers from multilayer film A1 with the STRAP-A process.

Experiment	Initial film mass [g]	Polymer yield [wt %] PE ^[a]	EVOH	PET	Overall mass balance [%]
1	38.30	4.98	2.61	94.15	101.74
2	38.28	5.33	3.26	95.03	103.62
3	38.35	5.42	3.14	93.81	102.37
average	38.31	5.24	3.01	94.33	102.58
STDEV	0.04	0.23	0.34	0.62	0.96

[a] PE solids contain the EVA component.

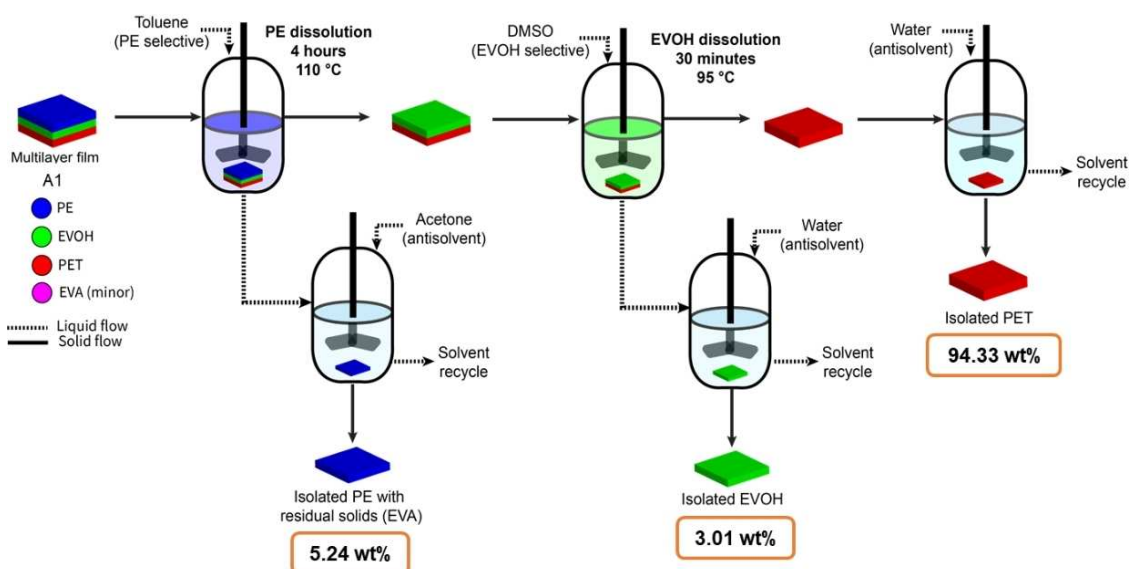


Figure 2. STRAP-A process schematic for the separation and recovery of the polymer components in a multilayer film A1 manufactured by Amcor.

precipitate the polymer as the temperature is decreased. In this way, we can have a constant solvent composition throughout the process that targets a specific polymer component in the multilayer film. The thermodynamic computational tools allowed us to quickly screen potential solvent compositions for the recovery of EVOH and PE using this approach.

Computational modeling results

COSMO-RS predictions were used to identify potential solvent compositions for the EVOH recovery, leveraging the ability of this method to capture temperature-dependent solubilities. Figure 3 shows COSMO-RS predictions of EVOH solubility as a function of temperature and DMSO-water compositions. For comparison, the predicted EVOH solubility in pure DMSO at 95 °C is 22.77 wt%, which is similar to the experimentally measured solubility of 24.02 wt%. This comparison supports the validity of COSMO-RS to predict EVOH solubilities. The STRAP-A process dissolves EVOH in pure DMSO at 95 °C. Water is then added as the antisolvent to achieve a final 20% DMSO-80% water (v/v) mixture at a lower temperature, reducing the EVOH solubility to trigger its precipitation. These steps are indicated by the labeled white arrow in Figure 3. This first process takes advantage of the large solubility difference (as predicted by COSMO-RS and measured experimentally) between the two states to achieve a high EVOH recovery.

Figure 3 indicates that the STRAP-B process could achieve the same decrease in solubility to precipitate the polymer by reducing the temperature without varying the solvent composition. The COSMO-RS solubility calculations indicate that EVOH recovery should be possible in pure DMSO, which is consistent with the inability of pure DMSO to dissolve EVOH at room temperature. However, the experimental recovery of EVOH upon cooling was low. Therefore, different mixture composi-

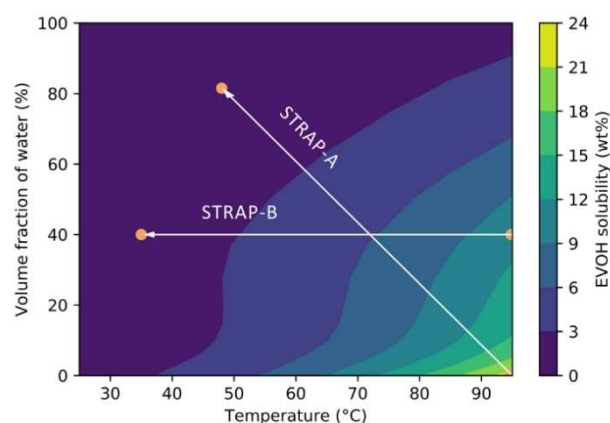


Figure 3. COSMO-RS solubility predictions for EVOH in DMSO-water mixtures. Two EVOH recovery processes are shown by arrows. The STRAP-A process dissolves EVOH in pure DMSO at 95 °C (predicted solubility 22.77 wt%) and precipitates the EVOH in 81.5% water at 48 °C (predicted solubility 1.23 wt%). The STRAP-B process dissolves EVOH in 40% water at 95 °C (predicted solubility 10.45 wt%) and precipitates the EVOH by reducing the temperature to 35 °C (predicted solubility 1.72 wt%).

tions were tested. A 60% DMSO-40% water mixture (v/v) was selected for EVOH recovery since it was predicted to have a satisfactory solubility (10.45 wt%) at 95 °C and low solubility (1.72 wt%) at 35 °C; the latter value is comparable to the predicted EVOH solubility for the final step of the STRAP-A process. Experimental values indicated a solubility of 7.17 wt% at 95 °C and 0.01 wt% at 35 °C, leading to a high EVOH recovery. We note that COSMO-RS overpredicts the experimental solubility, but general trends are correctly captured.

Similar COSMO-RS solubility predictions were computed for PE in various toluene-acetone mixtures to identify if any mixtures were suitable for recovery. The experimental solubility of PE in pure toluene (14.56 wt% at 110 °C) was used as a

reference input to the COSMO-RS calculations. The computational results indicate that PE solubility in 90% toluene-10% acetone (v/v) is 0.02 wt% at 55 °C. This result suggests that adding only a small fraction of acetone to toluene makes PE insoluble in the mixture, even at elevated temperatures; similar results were obtained for other volume fractions of acetone. This low solubility was verified experimentally, where none of the tested toluene-acetone mixtures could dissolve the PE. Therefore, pure toluene was still used in order to ensure initial PE dissolution. Pure toluene still allowed the PE precipitation to occur with a decrease in temperature.

STRAP-B with multilayer film A1

The STRAP-B process was completed with the solvent compositions allowing for a temperature induced precipitation of the EVOH and PE polymers using the thermodynamic insights from the former section. Table 2 shows that similar polymer yields to STRAP-A were obtained for PE, EVOH, and PET with STRAP-B, demonstrating that temperature-controlled dissolution and precipitation for the polymer recovery is a promising approach that could reduce costs and make solvent-based recycling processes more attractive.

One of the major differences between the STRAP processes was in the PE yield achieved. The PE yield was higher with STRAP-A where acetone was used as an antisolvent to precipitate the polymer. This difference in yield is due to the presence of ethyl vinyl acetate (EVA), which is a minor component in the multilayer film A1. The EVA is soluble in toluene and can also precipitate when acetone is added, along with the PE. An *N*-methylpyrrolidinone (NMP) step was previously considered for the EVA removal from the PE but we found that this separation can be easily done by decreasing the temperature of the toluene to precipitate PE, as EVA still remains in the solvent. The EVA can then be precipitated with the addition of acetone as an antisolvent. Figure 4 shows the solvents, temperatures and dissolution times that were used in each process step for STRAP-B.

ATR-FTIR characterization of polymers recovered with STRAP-A and STRAP-B

Previous Fourier-transform (FT)IR spectroscopy results have shown that the PE, EVOH, and PET recovered from physical mixtures of these components and from the multilayer film A1 using the STRAP-A process were indistinguishable from the corresponding virgin resins.^[11] The FTIR spectra of the PE, EVOH, and PET recovered with STRAP-A and STRAP-B were for the most part similar, indicating that pure polymers can be obtained using either precipitation method after selective dissolution (Figure 5). One of the major differences was observed in the PE spectra, where the polymer recovered through STRAP-A had extra peaks around 1736, 1240, 1097, and 1018 cm⁻¹ (red spectrum in Figure 5a). These peaks were previously attributed to EVA. It was observed in the spectra of the PE recovered through STRAP-B that the EVA peaks were less prominent, confirming that through this process the EVA polymer can be separated from the PE. Further removal of the EVA from the solvent required the addition of acetone as antisolvent. The solids recovered from that additional step were characterized and the spectra confirmed mainly EVA peaks (Figure 6b). In the PET recovered from STRAP-A and B, there were low intensity peaks around 3000 cm⁻¹ possibly coming from PE remnants in the PET (Figure 5c). The PET purification could be improved by changing solvent amounts and dissolution times of the other polymers. A quantitative measurement of the amount of PE present in the PET could be completed to determine if the PE is significant enough to affect the PET properties.

STRAP-C with multilayer film A2

Solvent screening for PETG component in multilayer film A2

An A2 multilayer film, manufactured by Amcor, was used to demonstrate the STRAP process with a more complex feed composition. This film included PETG as an additional polymer component. If the same STRAP process developed for film A1 were applied to film A2, both PETG and PE would be dissolved by toluene. Therefore, we used the computational approach to identify alternative solvents capable of dissolving PETG to develop a new STRAP process (referred to as STRAP-C). We first obtained HSPs for 850 solvents and calculated each solvent's

Table 2. Yield of polymers from multilayer film A1 with the STRAP-B process.^[a]

Experiment	Initial film mass [g]	Polymer yield [wt %]				Overall mass balance [%]
		PE	EVOH	EVA	PET	
1	38.272	4.02	3.16	1.06	93.47	101.86
2	38.34	4.05	3.27	0.76	93.81	102.84
3	38.26	4.53	3.24	0.78	92.69	102.02
average	38.29	4.20	3.22	0.91	93.32	102.24
STDEV	0.04	0.28	0.06	0.17	0.57	0.53

[a] Polymers were precipitated by reducing the solvent temperature. A DMSO 60% – water 40% solvent mixture was used for EVOH and toluene was used for PE. More experimental details can be found in the Supporting Information.

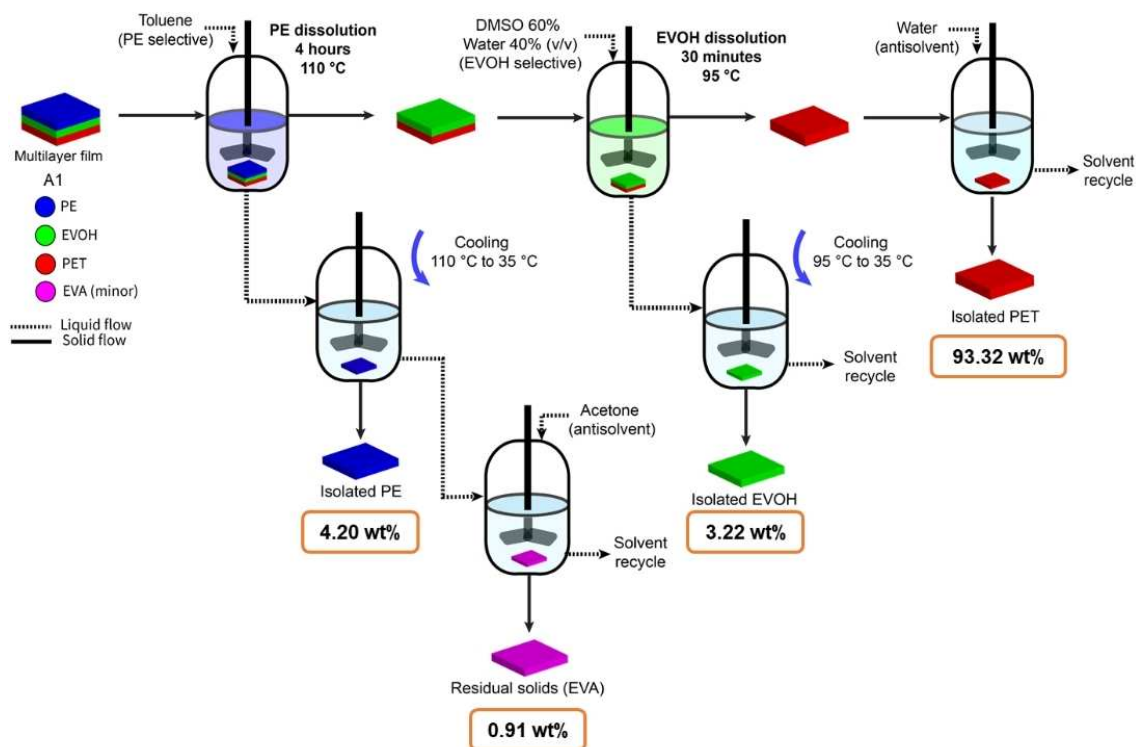


Figure 4. STRAP-B process schematic for the separation and recovery of the polymer components in a multilayer film A1 manufactured by Amcor.

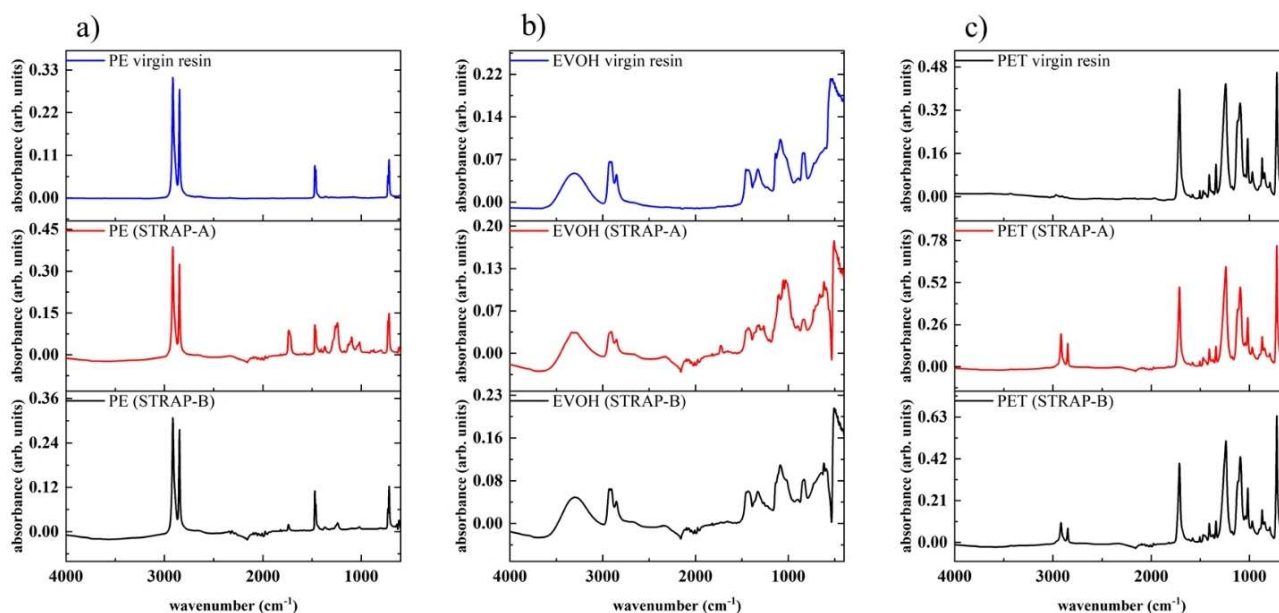


Figure 5. Attenuated total reflectance (ATR)-FTIR spectra of the virgin resins and polymers recovered from multilayer film A1 through STRAP-A and STRAP-B: (a) PE, (b) EVOH, (c) PET.

distance in HSP space (R_a) to the HSPs of PETG. Each value of R_a was then normalized by the radius of the PETG solubility sphere (R_0). Only solvents with values of R_a/R_0 less than 1 are expected to dissolve PETG. Table 3 includes HSP values for common, readily available solvents. Toluene, cyclohexanone, 1,4-dioxane,

and triethylamine are recognized as good candidates with R_a/R_0 values less than 0.7. THF, whereas 1,1-dichloroethane and ethyl acetate have R_a/R_0 values between 0.9 and 1, indicating probable dissolution. DMF and ethanol are identified as poor solvents with R_a/R_0 values greater than 2. However, PETG is a

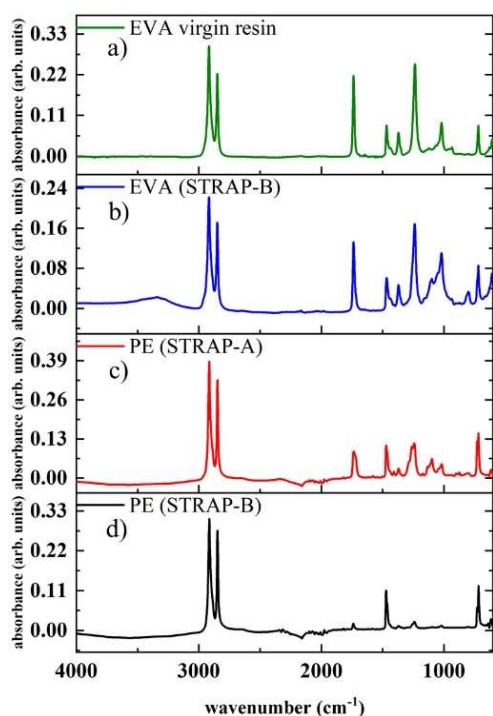


Figure 6. ATR-FTIR spectra: (a) EVA virgin resin, (b) EVA recovered through STRAP-B from multilayer film A1, (c) PE recovered through STRAP-A from multilayer film A1, (d) PE recovered through STRAP-B from multilayer film A1.

copolymer that is typically synthesized from terephthalic acid (TPA), ethylene glycol (EG), and 1,4-cyclohexanedimethanol (CHDM), with the ratio of EG/CHDM influencing its properties.^[34] Unfortunately, available HSP values for PETG do not provide any information on its composition;^[26] consequently, we also performed COSMO-RS solubility calculations, which consider both the structure and composition of this copolymer, to identify solvents capable of dissolving PETG. In this case, PETG was modeled as a random copolymer with a 2:1 molar ratio of EG/CHDM based on known information on the PETG component in film A2. Table 3 shows that the HSP and COSMO-RS results agree with each other for toluene, cyclohexanone, 1,4-dioxane, THF, and ethanol; these solvents have large COSMO-RS predicted solubilities and low values of R_a/R_0 . However, the COSMO-RS results also identify DMF as a potential good solvent. Based on these results, toluene, cyclohexanone,

dioxane, DMF, and THF could all be good solvents for PETG dissolution. Toluene was eliminated as a possible solvent for PETG because it also dissolves PE. We also eliminated dioxane as a solvent due to potential health hazards and eliminated cyclohexanone because ketones can be unstable. We thus selected THF and DMF as possible solvents for further experimental investigation.

After the solvent selection for the PETG component, the STRAP-C process was applied experimentally to the A2 multilayer film. The PETG dissolution was done at 65 °C for 4 h using THF, before any other polymer component. Separating the PETG entirely from the film using THF was difficult, as some polymer was left behind after the dissolution time was completed. This was not improved with different polymer/solvent ratios or a longer dissolution time. This separation inefficiency was attributed to the film strips being glued together possibly due to the presence of THF. This behavior was not observed whenever DMF was used for the PETG separation. Since DMF also dissolved the EVOH component, different THF/DMF mixtures were tested experimentally to decrease the EVOH solubility because THF is a poor solvent for EVOH (R_a/R_0 is 1.34 and the COSMO-RS predicted solubility is 1.78 wt%). A 40% THF-60% DMF (v/v) solvent mixture was determined to selectively dissolve PETG and not EVOH. This mixture was used for the actual film and the PETG yields improved from 15.28 to 20.83 wt%. 1-Propanol was selected as an antisolvent to precipitate the PETG since it had a low solubility in that solvent and it allowed for an easier solvent recycle after the recovery of PETG. Future work could consider solvent compositions to dissolve and precipitate the PETG component without the addition of an antisolvent. The STRAP-C results are presented in Table 4. For the remaining polymer components, which were PE, EVOH, PET, and EVA, the steps from STRAP-B were used, as shown in Figure 7.

ATR-FTIR characterization of polymers recovered with STRAP-C

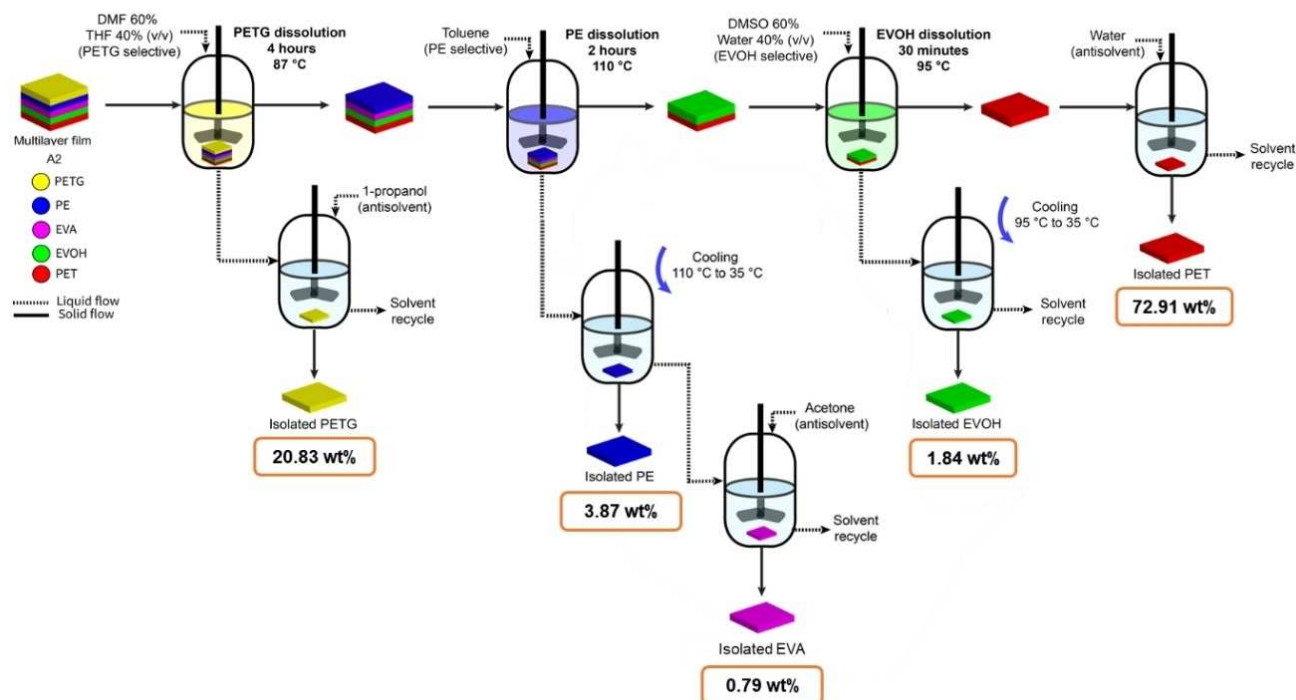
The FTIR spectra of the PETG, EVOH, EVA, and PET recovered using STRAP-C looked mostly similar to the corresponding virgin resins (Figure 8). Some notable differences were observed in the PE spectra, where additional peaks were present around 1740, 1370, 1240, and 1020 cm^{-1} . These could be due to PETG

Table 3. COSMO-RS and HSP solubility predictions for PETG.

Solvent	COSMO-RS predicted solubility [wt%]	HSP values (R_a/R_0)	Boiling point [°C]	Temperature for COSMO-RS prediction [°C]
toluene	47.82	0.43	110.6	110
cyclohexanone	36.80	0.58	155.4	100
1,4-dioxane	35.80	0.69	101.2	100
DMF	14.41	2.17	152.8	100
THF	10.21	0.90	66	65
1,1-dichloroethane	1.43	0.96	56.3	55
1-propanol	1.02	2.42	97.2	96
triethylamine	0.19	0.66	88.8	85
ethanol	0.15	2.84	78.2	78
ethyl acetate	0.04	0.98	77.1	75

Table 4. Yield of polymers from a multilayer film A2 with the STRAP-C process.

Experiment	Initial film mass [g]	Polymer yield [wt %]		EVOH	EVA	PET	Overall mass balance [%]
		PETG	PE				
1	38.61	21.04	3.74	2.05	0.0031	73.11	99.94 %
2	38.53	21.00	3.48	1.93	0.99	73.25	100.65 %
3	38.66	20.44	4.39	1.55	0.59	72.39	99.35 %
average	38.60	20.83	3.87	1.84	0.79	72.91	99.98 %
STDEV	0.07	0.34	0.47	0.26	0.28	0.46	0.65 %

**Figure 7.** STRAP-C process schematic for the separation and recovery of the polymer components in a multilayer film A2 manufactured by Amcor.

remnants that were not separated in the first dissolution step. A quantitative measurement would be needed to determine if the PETG amounts in the PE could affect its properties. Dissolution times and solvent amounts could be further adjusted to improve the purity of the PE and remove more of the PETG.

Technoeconomic analysis

A TEA was completed to determine if it is possible to use any of the discussed STRAP processes to produce recycled polymers at a price that is comparable to the virgin resins. The minimum selling price (MSP) of the recycled resins was calculated for a process that operated continuously for 20 years with a 10% discount rate and a free input stream of the multilayer film (using a discounted cash flow analysis). We developed the STRAP process models in Aspen Plus (V11 Aspen Technology)

based on our reported experimental mass balances. We estimated the size and cost of the required equipment in the three different STRAP processes with the multilayer films A1 and A2. The equipment purchase prices and variable operating costs of the distillation columns, pumps and heat exchangers were estimated using Aspen Process Economic Analyzer (V11 Aspen Technology), whereas the other equipment prices were estimated using engineering methods such as the proportional expression and the power law rules based on the cost data in our previous study.^[11]

We calculated the MSPs of the STRAP-A, STRAP-B, and STRAP-C processes based on the material balances and economic parameters given in Tables S6 and S7 in the Supporting Information. The total capital investment (in millions) was \$25.65 for STRAP-A, \$22.42 for STRAP-B, and \$31.78 for STRAP-C. The MSPs of the STRAP-A and STRAP-B processes with the multilayer film A1 were calculated to be

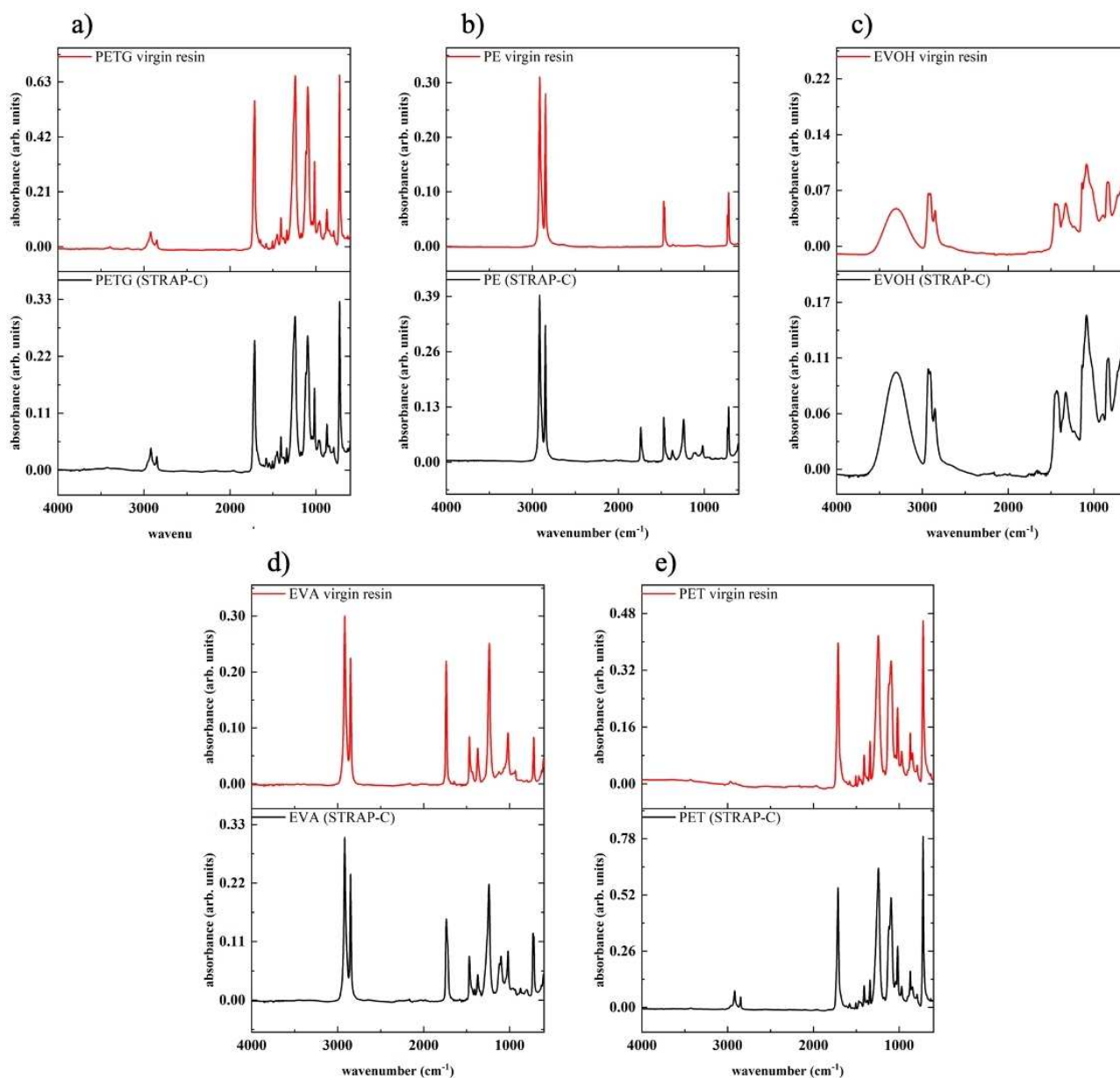


Figure 8. ATR-FTIR spectra of the virgin resins and polymers recovered from multilayer film A2 through STRAP-C: (a) PETG, (b) PE, (c) EVOH, (d) EVA, and (e) PET.

$\$2.05 \text{ kg}^{-1}$ and $\$1.62 \text{ kg}^{-1}$, respectively, at a feed rate of 3000 tonnes per year, as shown in Figure 9. According to our estimates, the MSP of the STRAP-B process was 21.0% lower than that of the STRAP-A process. This is because STRAP-B produced similar polymer yields to STRAP-A but without the distillation-based separation of the solvents and antisolvents. Even with the additional EVA recovery step in STRAP-B, it still has a lower MSP than STRAP-A. Furthermore, the MSP of STRAP-A is higher than that of the previous reported STRAP process^[11] since the proposed STRAP processes in this paper include extruders for each of the recycled polymers (Figures S2–S4). The extruder accounts for 55.22% of the total equipment purchase cost for STRAP-A.

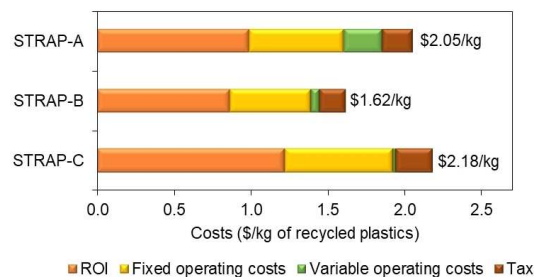


Figure 9. MSP and breakdown of revenue destinations for STRAP-A, STRAP-B, and STRAP-C.

The STRAP-C process was estimated to have an MSP of $\$2.18 \text{ kg}^{-1}$ at the feed rate of 3000 tonnes per year with multilayer film A2. The higher MSP is explained by the fact that STRAP-C recovers an additional polymer, PETG, where DMF/THF and 1-propanol are used as the solvent/antisolvent pair and distillation steps are required to recover and recycle the solvents. In general, by selecting low-boiling solvents and minimizing the amount used, the utility cost of any required distillation columns can be decreased. As a result, even though STRAP-C requires higher capital investment, it operates at a lower variable operating cost.

We also conducted a feed rate sensitivity analysis for the STRAP-B and STRAP-C processes in order to evaluate effects of economies of scale. The results, illustrated in Figure 10, indicate that, between 1000–30000 tonnes per year of feed, the capital investment appears to increase while the MSP begins to flatten out. The MSP of the STRAP-C process is lower than that of the STRAP-A counterpart as the feed capacity increases, especially after approximately 7000 tonnes per year of feed.^[11] This is because, at 3000 tonnes per year, the annual capital investment of the STRAP-C process is 62.58% of the total required cost, whereas the annual capital investment accounts for 53.81% of the total required cost of the STRAP-A process. For example, at 7000 tonnes per year of feed, the MSPs of the STRAP-A and STRAP-C processes are $\$1.43 \text{ kg}^{-1}$ and $\$1.40 \text{ kg}^{-1}$, respectively. As the scale increases especially after 15000 tonnes per year, the MSPs of the recycled polymers (STRAP-A: $\$1.07 \text{ kg}^{-1}$, STRAP-B: $\$0.78 \text{ kg}^{-1}$, STRAP-C: $\$0.95 \text{ kg}^{-1}$) are comparable to the average market values of the virgin resins which has been around $\$0.9$ – $\$1.2 \text{ kg}^{-1}$.^[12,35,36] These estimates show that the STRAP process

could be implemented at a large scale to recover and recycle the components in different post-industrial multilayer films.

An economic analysis of the recycled polymers from the STRAP-C process was conducted by changing the polymer selling price from 0.8 to $\$3.0 \text{ kg}^{-1}$ for the three different feed capacities, as shown in Figure 11. As expected, the internal rate of return (IRR) of the STRAP-C process increases with increasing the selling price. At the polymer selling price of $\$3.0 \text{ kg}^{-1}$, the IRR of the STRAP-C process is estimated to be 16.63, 32.51, and 40.99% for 3000, 9000, and 15000 tonnes per year, respectively. The economic feasibility of the STRAP process shows to be improved with the larger feed capacity due to the effect of economies of scale. Furthermore, to investigate the parameters which have impacts on the economics of the STRAP-C process, we conducted a sensitivity analysis by changing in $\pm 30\%$ of the selected parameters (Figure 12). The MSP of the STRAP-C is sensitive to the extruder equipment cost and the project lifetime. This is because the cost structure of the STRAP process is dominated by the capital investment. It is also observed that the solvent amount to the polymer influences the MSP since the amount of solvents has an effect on the equipment sizes. Overall, the TEA results for STRAP-B and STRAP-C demonstrate that the STRAP process can remain economically viable as the plastic feed complexity increases. By using solvents that enable temperature-controlled dissolution and precipitation, the capital costs can be significantly reduced by decreasing the required number of separation equipment for solvent recovery.

Conclusions

In this work, the solvent-targeted recovery and precipitation (STRAP) technology was demonstrated to separate and recover the constituent polymers in two different rigid post-industrial multilayer plastic films manufactured by Amcor. The thermodynamic computational techniques permit initial predictions of polymer solubility in different solvents and solvent mixtures, reducing the number of experiments needed. It was previously determined that the main cost drivers in the STRAP process were the required distillation steps for the recycle of solvents and antisolvents. We show in this study that reducing the use

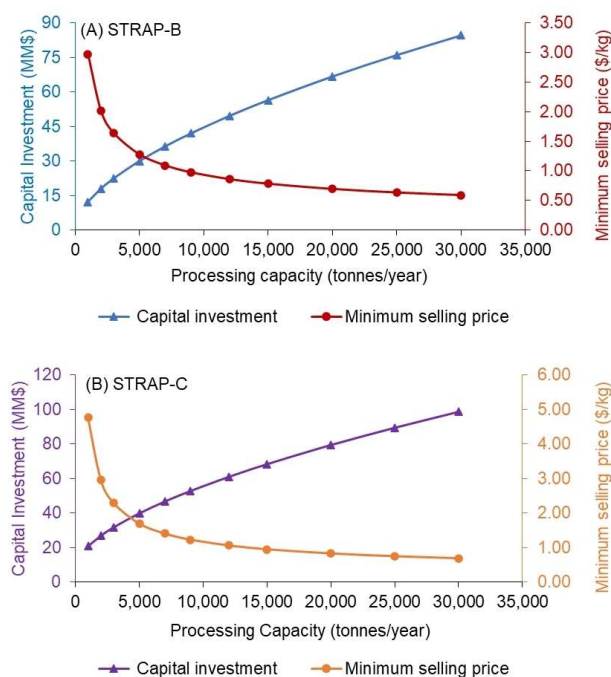


Figure 10. Total capital investment and MSP for (a) the STRAP-B process and (b) the STRAP-C process as a function of process feed capacity.

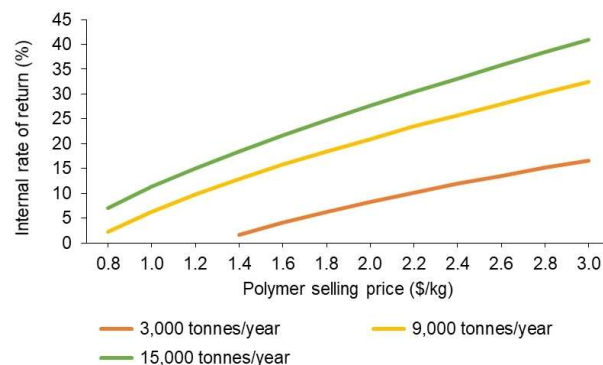


Figure 11. Internal rate of return for the STRAP-C process as function of polymer selling price for three different feed capacities.

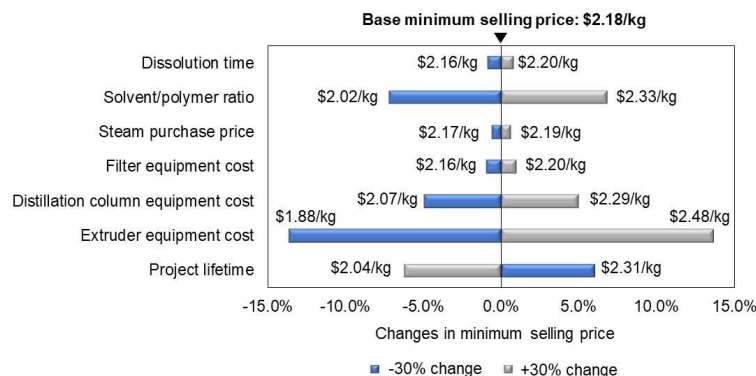


Figure 12. Sensitivity analysis ($\pm 30\%$) of the parameters for the STRAP-C process.

of antisolvents and replacing them with solvents or solvent mixtures that enable a temperature-controlled polymer dissolution and precipitation is an alternative to make STRAP more economically competitive. This was observed with our STRAP-B process which produced similar yields and was more efficient in the polymer separation than STRAP-A, while using less amounts of antisolvents. In addition to this, we have demonstrated that the STRAP process can remain economically competitive even as the plastic feed complexity increases, as was demonstrated in STRAP-C with multilayer film A2. Solvent mixtures are of great importance in solvent-based recycling systems and tuning the solvent composition will be a valuable parameter to obtain the desired polymer yields. Further developing the computational tools to predict polymer solubilities in different solvents with varying compositions will support in implementing this approach for the recycling of multilayer plastic films. Dissolution times and solvent amounts can be further adjusted to improve the purity of some of the recovered polymers using STRAP. Quantitative measurements could be done to assess if some impurities in the recovered polymers would affect their properties. Future work will also focus on using STRAP to process multilayer films with polymers not reported in this study, like polyamides, polystyrene, polypropylene, and polyvinyl chloride. Ideally, this would include finding solvents or solvent combinations that facilitate a thermally driven dissolution and precipitation of each polymer. Moreover, we will consider the removal of additives like adhesives and inks, which can also be present in plastic films. Our computational modeling framework will give us an understanding of the thermodynamic polymer-solvent properties and will allow for the design of solvent systems that can be used to efficiently separate and recover the polymer components in different multilayer plastic films.

Experimental Section

Experimental procedure with multilayer film A1

The main goal of the STRAP process is to separate the constituent polymers in a multilayer plastic film using a series of solvent washes. A model film manufactured by Amcor was used to develop and demonstrate the experimental procedure. The plastic film was an A1 film composed primarily of PE, PET, EVA, and EVOH. Before a typical STRAP experiment, the multilayer film was cut into $1 \times 1 \text{ cm}^2$ stamps. A 250 mL round-bottom flask connected to a reflux condenser with a cold-water supply line was used. The round bottom flask, which would contain the corresponding solvent and the pieces of multilayer film in 40 g batches, was partially submerged in a 1500 mL dish containing silicone oil as a heat transfer fluid. The system was heated to the desired dissolution temperature with an electric heat plate equipped with a magnetic stir drive and the stirring rate was adjusted to have a constant mixing. Based on the computational analysis for solvent selection, we experimentally assessed the recovery of each polymer in the multilayer film A1 by performing three processing steps: (1) selectively dissolving the PE fraction in toluene at 110°C for 4 h, then separating the solubilized fraction from the EVOH and PET via mechanical filtration; (2) selectively dissolving the EVOH fraction in DMSO at 95°C for 30 min, then separating the solubilized fraction from the remaining PET via mechanical filtration; and (3) precipitating the solubilized PE and EVOH fractions with the addition of acetone or water, respectively. The precipitated PE and EVOH were then separated from the toluene-acetone and DMSO-water mixtures by filtration. The resulting solid fractions were dried in a vacuum oven overnight at 65°C to remove residual solvents. This method was identified as STRAP-A, where the addition of antisolvents is required to precipitate the target polymers.

An alternative method for the precipitation of the dissolved polymers was considered. This consisted of lowering the temperature of the solvent and the dissolved polymer to 35°C . For this method, labeled as STRAP-B, solvent mixtures were considered to both dissolve the target polymer at high temperatures and precipitate the polymer at lower temperatures, reducing the need for high amounts of antisolvents to be added after the dissolution to precipitate the polymer. Toluene-acetone and DMSO-water mixtures were tested for the recovery of PE and EVOH, respectively.

Experimental procedure with multilayer film A2

The same experimental set up was used for the multilayer film A2, which consisted of PETG, PE, EVOH, PET, and EVA. Similarly to A1, the multilayer film was cut into $1 \times 1 \text{ cm}^2$ stamps. We experimentally assessed the recovery of each polymer in the multilayer film A2 by performing four processing steps: (1) selectively dissolving the PETG fraction in a mixture of 60% DMF-40% THF (v/v) at 87 °C for 4 h, then separating the solubilized fraction from the PE, EVOH, PET, and EVA via mechanical filtration; (2) selectively dissolving the PE and EVA fraction in toluene at 110 °C for 2 h, then separating the solubilized fraction from the EVOH and PET via mechanical filtration; (3) selectively dissolving the EVOH fraction in a mixture of 60% DMSO-40% water (v/v) at 95 °C for 30 min, then separating the solubilized fraction from the remaining PET via mechanical filtration; and (4) precipitating the solubilized PETG by adding 1-propanol as antisolvent and precipitating the PE and EVOH fractions by reducing the temperature of the solvents containing the dissolved polymers to 35 °C. The precipitated PETG, PE, and EVOH solids were then separated from the corresponding solvents by filtration. The EVA was recovered by adding acetone as antisolvent to the toluene after removing the PE solids. The resulting five solid fractions were dried in a vacuum oven overnight at 65 °C to remove residual solvents. The process with multilayer film A2 was labeled as STRAP-C.

Characterization of solids from the STRAP process

ATR-FTIR spectroscopy was used to characterize the separated polymer fractions from the STRAP process and compare their spectra to virgin resins. The instrument was a Bruker Vertex 70 with a liquid nitrogen-cooled Mercury-Cadmium-Telluride (MCT) detector. The ATR cell used was a MIRacle single reflection cell equipped with a diamond crystal (Pike Technologies). In a typical measurement, 128 scans were averaged with a 4 cm^{-1} resolution and range from 4000–400 cm^{-1} .

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: molecular simulation · plastics recycling · polymer solubility · process design · process economics

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